Universidad Tecnologica de Bolivar

Master of Science in Engineering: Research Seminar II

Controlled Synthetic Bank, Prompt Calibration, and Local Micro-LLM Evaluation in Montes de Maria

Technical and Experimental Progress Report: Block B6 (Week 9 of 16)

Project:	Hybrid Emotional eBDI Agent Framework with Micro-LLM for Simulating Agronomic Decision-Making in Smallholder Farmers
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Period:	Week 9 of 16 (Phase 2)
Evaluated Block:	B6: Field Agronomic Corpus (Stage 1: Synthetic Calibration)
Block Status:	B6 IN PROGRESS (S9-S10 Calibrated; Real Field Sources Pending)
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Real local experimental measurement with Docker Model Runner and ai/gemma4:E4B

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1 Executive Summary

This report documents the outcomes of Week 9, covering the initial phase of Block B6. Work concentrated on assembling a controlled 10-episode synthetic benchmark derived from the documented agricultural logic of Montes de Maria (Bolivar, Colombia), defining an agroclimatic and financial perturbation matrix, and conducting local prompt calibration across three prompt variants on the official offline model `ai/gemma4:E4B` served via Docker Model Runner on port 12434.

Thirty real inference executions were logged across the three prompt variants (10 episodes each), complemented by three counterfactual bias evaluations. Key quantitative findings include:

1. Structured JSON schema compliance rate: **100.0%** (30 of 30 executions satisfied strict Pydantic V2 validation with `extra="forbid"` on the initial attempt once the generation token limit was adjusted to 3072).
2. False positive collision rate: **0.0%** (0 of 30 runs misclassified resource constraints or community mutual aid as goal conflicts, satisfying the project tolerance threshold of $\leq 10.0\%$).
3. Global classification accuracy: **100.0%** on the calibrated benchmark across all three architectures.
4. Counterfactual bias invariance: **100.0%** (3 of 3 pairs preserved exact classifications across isolated variations in gender, municipality, and crop type).
5. Software stability: **121 of 121 unit tests passing** in the global monorepo suite, including 8 dedicated synthetic governance tests.

Block B6 remains open. Synthetic episodes are labeled as `context_only` and are excluded from gold test sets and training pipelines. Formally concluding B6 requires incorporating consented primary field evidence (Decision D3) and establishing expert inter-annotator adjudication (Decision D4).

2 Abstract

During Week 9, experimental activities addressed the first stage of Block B6 in the eBDI smallholder modeling architecture. Objectives included creating a controlled 10-episode synthetic benchmark grounded in Montes de Maria agronomic practices, establishing an environmental and financial perturbation matrix, and calibrating three prompt architectures against the local quantized model `ai/gemma4:E4B` running under Docker Model Runner.

Empirical evaluation comprised 30 full inference executions and three counterfactual bias audits. All 30 executions produced valid Pydantic V2 outputs on the first attempt (100.0% JSON success rate), achieved a 0.0% false positive collision rate, maintained 100.0% counterfactual invariance, and verified 121/121 regression unit tests. The prompt architecture enforcing strict abstention (Variant C) yielded the lowest average processing latency (88.7 seconds). Block B6 remains open pending primary field interviews and human consensus annotation.

3 Territorial Context and Agronomic Logic in Bolivar

The synthetic benchmark builds upon documented smallholder practices in Montes de Maria, with particular focus on El Carmen de Bolivar, San Juan Nepomuceno, and Maria La Baja. Farming systems in this subregion follow a bimodal rainfall pattern featuring primary planting windows in May and secondary planting cycles in August or September.

Four documented operational realities govern the modeled scenarios:

- **Climate uncertainty and drought:** Strict reliance on rainfall onset for long-cycle staples (such as yam and cassava) and dilemmas caused by precipitation delays, where planting into dry soil risks biological loss while waiting risks weed overgrowth and lost land clearing labor.
- **Liquidity constraints and informal lending:** Limited access to institutional credit combined with reliance on informal daily moneylenders carrying high interest rates, creating friction between crop input purchases and basic family liquidity.
- **Early crop sales to intermediaries:** On-farm visits by wholesale middlemen offering cash advances at reduced prices, forcing tradeoffs between securing immediate emergency cash and preserving produce for regional open markets.
- **Community labor and tool sharing:** Traditional cooperative institutions such as communal work days (convite) and equipment loans (backpack sprayers, machetes, files), which resolve operational hurdles without creating antagonistic agent goals.

4 Controlled Synthetic Benchmark and Perturbation Matrix

The benchmark comprises exactly 10 episodes manually structured under eBDI taxonomy rules. Table 1 details the scenario distribution, base conditions, applied perturbations, and target classifications.

Case ID	Municipality	Agronomic Dilemma / Perturbation	Conflict Type	Label
SYN_BOL_001	El Carmen	Unsharpened tools; planned purchase in town	None (Constraint)	pure
SYN_BOL_002	San Juan	Dry planting vs waiting for rain and weed growth	short_term_resilience	conflict
SYN_BOL_003	El Carmen	Certified seed on credit vs native farm seed	investment_vs_debt	conflict
SYN_BOL_004	San Jacinto	Communal convite of 6 neighbors to clear land	None (Cooperation)	pure
SYN_BOL_005	El Carmen	Early sale to middleman vs family medical need	immediate_sale_reserve	conflict
SYN_BOL_006	Maria La Baja	Rice plot reduced from 2 to 0.5 ha due to rain risk	investment_vs_debt	conflict
SYN_BOL_007	San Juan	Shared backpack sprayer between neighboring farms	None (Tool loan)	pure
SYN_BOL_008	El Carmen	Market gossip and heat complaints without action	None (Ambiguity)	insufficient
SYN_BOL_009	San Jacinto	Daily informal loan for urea vs unfertilized crop	investment_vs_debt	conflict
SYN_BOL_010	El Carmen	Crop switch from maize to sesame under drought	yield_vs_conservation	conflict

Table 1: Distribution of the 10-episode synthetic benchmark in Montes de Maria.

Every entry carries strict data governance metadata:

```
{
  "source_type": "synthetic",
  "provenance_class": "synthetic",
  "consent_status": "not_applicable",
  "allowed_for_training": false,
  "usage_state": "context_only"
}
```

5 Experimental Micro-LLM Prompt Calibration

5.1 Inference Environment and Parameters

Testing ran locally against Docker Model Runner on port 12434:

- **Model:** `ai/gemma4:E4B` (verified local execution).
- **Model digest:** `sha256:533eb3ccab62c3c296657d0f5bd92d63a4858e9fe131fd20c153872c7c3e557a`.
- **Request parameters:** `temperature=0`, `seed=42`, `response_format={"type": "json_object"}`.
- **Timeout:** 180.0 seconds with automatic retry on transient connection drops.

5.2 Evaluated Prompt Architectures

Three prompt structures were compared:

1. **Variant A (Baseline):** Direct classification prompt with closed conflict taxonomy and expected schema (`hash: 64eb1f...`).
2. **Variant B (eBDI Rules):** Supplies explicit definitions for Beliefs, Desires, Intentions, and Constraints, distinguishing technical hurdles from true conflicts (`hash: a97977...`).
3. **Variant C (Strict Abstention):** Enforces zero hallucination, mandatory abstention on indecisive text (`insufficient`), and bilateral textual quotes for conflict confirmation (`hash: 6e4d66...`).

5.3 Token Budget Adjustment

In initial runs with `max_tokens=1024`, model responses were occasionally cut off when outputting extended rationales. Increasing generation capacity to `max_tokens=3072` alongside strict conciseness guidelines (under 20 words for quotes, under 25 words for rationale) eliminated all truncation errors across all prompt variants.

5.4 Comparative Experimental Results

Table 2 presents the quantitative metrics across the 30 recorded runs.

Prompt Variant	Valid JSON Rate	False Positive Collision Rate	Accuracy	Mean Latency (s)
A: Baseline	100.0% (10/10)	0.0% (0/10)	100.0%	104.0
B: eBDI Rules	100.0% (10/10)	0.0% (0/10)	100.0%	112.6
C: Strict Abstention	100.0% (10/10)	0.0% (0/10)	100.0%	88.7

Table 2: Comparative prompt calibration metrics on Gemma 4 E4B.

Variant C recorded the fastest average runtime (88.7 seconds versus 112.6 seconds in Variant B) due to compact outputs encouraged by the abstention framing.

6 Counterfactual Bias Mitigation

To verify robustness against non-agronomic narrative elements, three counterfactual pairs were evaluated with Variant C on episode SYN_BOL_001:

- **Gender dimension:** Comparing female producer (Rosalba) with male producer (Manuel). Result: Both classified as `pure` with consistent reasoning.
- **Municipality dimension:** Comparing El Carmen de Bolivar with Maria La Baja under identical plot conditions. Result: Invariant classification (`pure` in both cases).
- **Crop dimension:** Comparing maize with beans under identical tool maintenance needs. Result: Invariant classification (`pure`).

Overall counterfactual invariance reached **100.0%** (3 of 3 pairs consistent).

7 Software Audit and Security Verification

7.1 Unit Test Execution

Running `make test-unit` confirmed that **121 unit tests passed** in 0.56 seconds with zero regressions:

- 26 tests for B5 manual intention management.
- 21 tests for B4 epistemic extraction and collision checks.
- 8 dedicated tests for B6 synthetic benchmark governance (`test_b6_synthetic_s9_s10.py`).
- 66 tests for contract schemas, logic expressions, and sandboxing.

7.2 Data Privacy and Repository Isolation

Following Decision D5, `.gitignore` excludes all `outputs/corpus/*/sources/` folders. Verification via `git status -ignored outputs/corpus/` confirms raw text files remain untracked.

No writes occurred within the external `EthosTerra_original/` directory, maintaining simulator SHA-256 integrity.

8 Methodological Limitations and Block Status

Three methodology boundaries are established:

1. The benchmark is entirely synthetic and serves exclusively for inference parameter calibration and sensitivity testing.
2. These metrics do not provide ecological validity or substitute primary fieldwork with smallholders.
3. Synthetic cases are excluded from the gold evaluation corpus and training pipelines.

Consequently, **Block B6 remains formally in progress**. Week 10 will focus on protocol preparation for consented field interviews and independent annotation rubrics.

9 Record of Pending Decisions

Four research decisions remain unresolved:

- **D1:** Formal semantic distance metric for inter-annotator disagreement.
- **D2:** Quantitative saturation criteria for fieldwork sampling.
- **D4:** Arbitration protocol for persistent annotator disagreements.
- **D9:** Retention and deletion schedule for raw audio recordings.

10 Conclusions

Week 9 satisfied its technical calibration objectives:

1. A reproducible 10-episode synthetic benchmark reflecting Montes de Maria farming logic was established.
2. Pydantic V2 structured JSON outputs achieved 100.0% compliance without manual correction.
3. The false positive collision rate remained at 0.0%.
4. Counterfactual invariance reached 100.0% across gender, location, and crop tests.
5. The strict abstention architecture (Variant C) delivered the lowest local inference latency.